Predicting Calorie Expenditure During Exercise

Using Regression-Based Machine Learning Approaches

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ABSTRACT

Accurate prediction of calorie expenditure during physical exercise is essential for personalized fitness planning, health monitoring, and the prevention of chronic diseases such as obesity and cardiovascular disorders. This project addresses the problem of predicting the number of calories burned during a workout session based on physiological and demographic attributes. We formulate this as a regression task and leverage the Kaggle Playground Series Season 5, Episode 5 dataset, which contains approximately 750,000 synthetically generated training samples with features including age, gender, height, weight, exercise duration, heart rate, and body temperature. We plan to evaluate and compare multiple regression models, including gradient boosting methods (XGBoost, LightGBM, CatBoost), tree-based ensembles, neural networks, and AutoML approaches, using Root Mean Squared Logarithmic Error (RMSLE) as the primary evaluation metric. Through systematic feature engineering, hyperparameter tuning, cross-validation, and model ensembling, this project aims to identify the most accurate and generalizable approach for calorie expenditure prediction.

1. INTRODUCTION

Physical inactivity and poor diet are among the leading risk factors for global mortality, contributing to conditions such as obesity, type 2 diabetes, and heart disease. A fundamental component of managing personal health is understanding energy expenditure specifically, how many calories an individual burns during exercise. While wearable devices and fitness trackers attempt to estimate this value, their accuracy varies significantly across devices and activity types. Developing reliable machine learning models that can predict calorie expenditure from easily measurable physiological parameters offers a scalable and cost-effective alternative.

The problem of calorie prediction is inherently challenging for several reasons. First, calorie expenditure depends on a complex interplay of factors: an individual's body composition, metabolic rate, exercise intensity, and physiological response (e.g., heart rate, body temperature) all play significant roles. Second, the relationship between these variables and calorie burn is nonlinear, a heavier individual exercising at a moderate heart rate may burn more calories than a lighter individual at the same heart rate, but the relationship is not simply proportional. Third, individual variability means that two people with similar demographic profiles may have very different energy expenditure patterns due to differences in fitness level, muscle mass, or genetic predisposition.

This project is important because accurate calorie prediction has broad applications across multiple domains. In personalized fitness and nutrition planning, reliable estimates help individuals set realistic dietary and exercise goals. In clinical settings, calorie expenditure models can support rehabilitation programs and weight management interventions. In the wearable technology industry, improved prediction algorithms can enhance the accuracy of consumer fitness devices. Furthermore, from a machine learning perspective, this problem serves as an excellent benchmark for comparing regression algorithms on tabular data with mixed feature types (categorical and continuous).

The key challenges we aim to address include:

1. Capture the nonlinear and interaction effects among features that drive calorie expenditure
2. Handle mixed data types, as the dataset includes both categorical and continuous features
3. Select an appropriate evaluation metric, the RMSLE metric penalizes under-predictions more heavily than over-predictions and is robust to outliers, making it well-suited for this problem
4. Ensure that models generalize well and do not overfit to the large training set

2. PROPOSED PROJECT

2.1. Problem Formulation

We formulate the calorie expenditure prediction task as a supervised regression problem. Given a set of input features describing a person's demographic profile (gender, age, height, weight) and exercise session characteristics (duration, heart rate, body temperature), the goal is to predict a continuous target variable: the number of calories burned. This is a regression task because the target variable (Calories) is continuous and unbounded, rather than belonging to a discrete set of categories.

2.2. Dataset Description

We will use the dataset from the Kaggle Playground Series Season 5, Episode 5 competition ("Predict Calorie Expenditure") [1]. This dataset was synthetically generated from a deep learning model trained on the original Calories Burnt Prediction dataset available on Kaggle [2]. The feature distributions are close to, but not identical to, the original data. The dataset is structured as follows:

Table 1: *Dataset Overview*

Component	**Details**
Training Set	~750,000 samples; features: Sex (male/female), Age, Height (cm), Weight (kg), Duration (min), Heart_Rate (bpm), Body_Temp (°C); target: Calories
Test Set	~250,000 samples with the same 7 features (no target variable)
Original Data	15,000 samples from the original dataset [2] with identical features; we may incorporate this as supplementary training data

We plan to perform the following preprocessing and feature engineering steps:

1. Encode the categorical Sex feature using label encoding or one-hot encoding
2. Check for and handle missing values or outliers
3. Create features such as BMI and (Duration * Heart rate) as an exercise intensity proxy
4. Apply log transformation to the target variable where appropriate given the RMSLE metric
5. Standardize or normalize continuous features as required by specific models.

2.3. Proposed Methodology

The central goal of this project is to systematically benchmark a diverse set of machine learning regression methods, compare their performance, and select the best-performing approach for calorie expenditure prediction.

We plan to implement the following models, organized by family:

Table 2: *Proposed Model Families*

Model Family	**Methods**	**Rationale**
Gradient Boosting	XGBoost [3], LightGBM [4], CatBoost, HistGradientBoosting	State-of-the-art on tabular regression benchmarks; captures nonlinear interactions
Tree-Based Ensembles	Random Forest [5], Extra Trees	Robust baselines that handle mixed features without explicit encoding
Neural Networks	Keras MLP	Explore deep learning on tabular data; complement tree-based approaches
AutoML	AutoGluon	Automated model selection and tuning as an additional benchmark

Each model will be evaluated under identical conditions using 5-fold cross-validation on the training set, ensuring a fair and reproducible comparison. The primary evaluation metric will be RMSLE, consistent with the Kaggle competition scoring. We will additionally report RMSE as a secondary metric for interpretability.

$$\text{RMSLE} = \left(\frac{1}{n}\sum_{i=1}^{n}(\log(1+\hat{y}_i) - \log(1+y_i))\right)^{\frac{1}{2}}$$

where:

- $\boldsymbol{n}$ is the total number of observations in the test set,
- $\boldsymbol{\hat{y}_i}$ is the predicted value of the target for instance
- $\boldsymbol{y_i}$ is the actual value of the target for instance
- $\boldsymbol{log}$ is the natural logarithm.

Hyperparameter tuning will be performed using Optuna with Tree-structured Parzen Estimation (TPE) sampling. Gradient boosting models will use early stopping to prevent overfitting while permitting large estimator counts. After benchmarking individual models, we plan to combine predictions using a stacking (stacked generalization) approach, where out-of-fold predictions from the base models serve as input features for a meta-learner (e.g., Ridge regression) that learns optimal blending weights. All results will be compiled into benchmarking tables and visualizations ranking models by RMSLE.

3. References

[1] Kaggle, "Predict Calorie Expenditure – Playground Series Season 5, Episode 5," 2025. Available: https://www.kaggle.com/competitions/playground-series-s5e5

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[3] T. Chen and C. Guestrin, "XGBoost: A scalable tree boosting system," in Proc. 22nd ACM SIGKDD Int. Conf. Knowledge Discovery and Data Mining, 2016, pp. 785–794.

[4] G. Ke et al., "LightGBM: A highly efficient gradient boosting decision tree," in Advances in Neural Information Processing Systems 30, 2017, pp. 3146–3154.

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